DATA INSPECTION AND ANALYSIS

1. Aim

To inspect and analyse a dataset using Python and Pandas by examining its dimensions, records, column names, data types, descriptive statistics, missing values, duplicate rows and categorical values.

2. Algorithm

1. Start the program.
2. Import the Pandas library.
3. Load the Excel dataset using **pd.read_excel()** and select the required worksheet.
4. Display the first five records using **head()** and the last five records using **tail()**.
5. Find the number of rows and columns using **shape**.
6. Display all column names using **columns** and inspect data types using **dtypes**.
7. Use **info()** to check non-null counts and dataset structure.
8. Use **describe(include='all')** to obtain summary statistics.
9. Check missing values with **isnull().sum()** and duplicate records with **duplicated().sum()**.
10. Count distinct values using **nunique()** and inspect category frequencies using **value_counts()**.
11. Display the results and stop the program.

3. Python Program

```
import pandas as pd

file_path = "socialmedia (1) (1) (1)(1).xlsx"
sheet_name = "socialmedia (1) (1) (1)"

df = pd.read_excel(file_path, sheet_name=sheet_name)

print("FIRST FIVE RECORDS")
print(df.head())

print("\\nLAST FIVE RECORDS")
print(df.tail())

print("\\nDATASET DIMENSIONS")
print("Rows and columns:", df.shape)
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])

print("\\nCOLUMN NAMES")
print(df.columns.tolist())

print("\\nDATA TYPES")
print(df.dtypes)

print("\\nDATASET INFORMATION")
df.info()

print("\\nDESCRIPTIVE STATISTICS")
print(df.describe(include="all").to_string())

print("\\nMISSING VALUES BY COLUMN")
print(df.isnull().sum())

print("\\nTOTAL MISSING VALUES")
print(df.isnull().sum().sum())

print("\\nDUPLICATE ROWS")
print(df.duplicated().sum())

print("\\nUNIQUE VALUES IN EACH COLUMN")
print(df.nunique())

print("\\nPLATFORM FREQUENCY")
print(df["Platform"].value_counts(dropna=False))
```

4. Output

The following output is generated from the selected Excel worksheet.

a) First five records

```
  UserID  Age  ... CurrentActivity ConnectionType
       1   56  ...       Commuting    Mobile Data
       2   46  ...       At school          Wi-Fi
       3   32  ...         At home    Mobile Data
       4   60  ...       Commuting    Mobile Data
       5   25  ...         At home    Mobile Data
```

b) Dataset dimensions

```
Rows and columns: (1000, 31)
Number of rows: 1000
Number of columns: 31
```

c) Column data types

```
UserID                      int64
Age                         int64
Gender                     object
Location                   object
Income                      int64
Debt                         bool
Owns Property                bool
Profession                 object
Demographics               object
Platform                   object
Total Time Spent            int64
Number of Sessions          int64
Video ID                    int64
Video Category             object
Video Length                int64
Engagement                  int64
Importance Score            int64
Time Spent On Video         int64
Number of Videos Watched    int64
Scroll Rate                 int64
Frequency                  object
ProductivityLoss            int64
Satisfaction                int64
Watch Reason               object
DeviceType                 object
OS                         object
Watch Time                 object
Self Control                int64
Addiction Level             int64
CurrentActivity            object
ConnectionType             object
```

d) Missing values by column

```
UserID                      0
Age                         0
Gender                      0
Location                    0
Income                      0
Debt                        0
Owns Property               0
Profession                  0
Demographics                0
Platform                    0
Total Time Spent            0
Number of Sessions          0
Video ID                    0
Video Category              0
Video Length                0
Engagement                  0
Importance Score            0
Time Spent On Video         0
Number of Videos Watched    0
Scroll Rate                 0
Frequency                   0
ProductivityLoss            0
Satisfaction                0
Watch Reason                0
DeviceType                  0
OS                          0
Watch Time                  0
Self Control                0
Addiction Level             0
CurrentActivity             0
ConnectionType              0
```

e) Number of duplicate rows

```
0
```

f) Platform frequency

```
Platform
TikTok       273
Instagram    256
YouTube      250
Facebook     221
```

5. Result

Thus, the dataset was successfully inspected and analysed using Pandas. Its records, dimensions, column names, data types, summary statistics, missing values, duplicate rows and category frequencies were examined through textual output without using visualisations.